for fuels of biogenic origins or made from captured carbon, therefore the ec and e ccu parameters should be taken into account and the S Fc and S Fccu values should be always 1.

5.4 The actual GHG intensity depends both on the properties of the fuel and on the

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SOURCE: MEPC.376(80).pdf
and e ccu parameters should be taken into account and the S Fc and S Fccu values should be always 1.

5.4 The actual GHG intensity depends both on the properties of the fuel and on the efficiency of the energy conversion. For CO2, the emission factors are based on the molar ratio of carbon to oxygen multiplied with the carbon mass of the fuel, assuming that all the carbon in the fuel is oxidized (stoichiometric combustion). The CH 4 and N 2O emissions factors are dependent on the combustion and/or conversion process in the energy converter.

5.5 For future use of, for example, fuel cells with a reforming unit, also electro -chemical reactions forming GHGs can be taken into account by this TtW methodology.

5.6 TtW default emission factors are provided in appendix 2 of these guidelines.

6 WELL-TO-WAKE (WtW)

6.1 The aim of the WtW methodology is to integrate WtT and TtW parts, to quantify the full life cycle emissions related to the production and use of a fuel.

6.2 The WtW GHG emission factor (gCO 2eq/MJLCV fuel or electricity) is calculated as follows:

Equation (3)

= +

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SOURCE: MEPC.376(80).pdf
6.2 The WtW GHG emission factor (gCO 2eq/MJLCV fuel or electricity) is calculated as follows:

Equation (3)

= +

where:

Term Units Explanation
GHGWtW gCO2eq/MJ(LCV) Total well-to-wake GHG emissions per energy unit from the use of the fuel or electricity in a consumer on board the ship
GHGWtT

gCO2eq/MJ(LCV)

Total well-to-tank GHG upstream emissions per energy unit of the fuel provided to the ship
GHGTtW

gCO2eq/MJ(LCV)

Total tank-to-wake GHG downstream emissions per energy unit from the use of the fuel or electricity in a consumer on board the ship

Equation (4)

= + + +
+ 1 ((1 1100 (_ + C)) × (2 × 2 + 4 × 4 + 2 × 2) +(1100 (+ C) × ×) × ×)

Note: terms , and are pending further methodological guidance to be developed by the Organization.

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SOURCE: MEPC.376(80).pdf
Note: terms , and are pending further methodological guidance to be developed by the Organization. For more details refer to section 5.2.

6.3 For the purpose of calculating WtW, the TtW value 2 as calculated in accordance with paragraph 5.3.2 should be used.

7 SUSTAINABILITY

7.1 The sustainability of marine fuels should be assessed considering the following themes/aspects on a life cycle basis:

.1 greenhouse gases (GHG);
.2 carbon source;
.3 source of electricity/energy;
.4 carbon stock direct land use change (DLUC);
.5 carbon stock indirect land use change (ILUC);
.6 water;
.7 air;
.8 soil;
.9 waste and chemicals; and
.10 conservation.

Other social and economic sustainability themes/aspects may be considered at a later stage.

7.2 The principle/objective in conjunction with the associated metrics/indicators of each of the sustainability theme/aspect are specified below.

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SOURCE: MEPC.376(80).pdf
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 14

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Table 1: Sustainability themes/aspects

Theme/aspect	Principle/Objective	Metric/Indicator
1. Greenhouse Gases (GHG)	Sustainable marine fuels generate lower GHG emissions than conventional marine fuels (energy-based weighted average of liquid petroleum products on 3 specific years of DCS data) on a life cycle basis.	1. GHG intensity in gCO2eq/MJ (GWP100); and GHG intensity in gCO2eq/MJ (GWP20) for comparative purposes.
2. Carbon source	Sustainable marine fuels do not increase GHG intensity from the use of fossil energy sources and the permanence of captured and stored	

carbon is ensured while also avoiding double counting across economic sectors.
1. Carbon source indicator, including its content (in %) and origin in feedstock used to produce final fuel product, i.e. Fossil, Biogenic, Captured Carbon (including direct air capture (DAC), point source fossil (PSF) and point source biogenic (PSB)), and Others (including mixture of

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SOURCE: MEPC.376(80).pdf
%) and origin in feedstock used to produce final fuel product, i.e. Fossil, Biogenic, Captured Carbon (including direct